Signal-to-Noise Ratios

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Introduction

Signal-to-noise (SNR) ratios in Taguchi's framework jointly summarise the mean and variance of a response. Three variants handle different quality targets: smaller-is-better, larger-is-better, and nominal-is-best. Maximising SNR selects factor combinations that produce both a desirable mean and low variance.

Prerequisites

Taguchi methods; variance and mean summaries.

Theory

- **Smaller-is-better** (minimisation): $\mathrm{SNR} = -10\log_{10}\left(\frac{1}{n}\sum y_i^2\right)$.
- **Larger-is-better** (maximisation): $\mathrm{SNR} = -10\log_{10}\left(\frac{1}{n}\sum 1/y_i^2\right)$.
- **Nominal-is-best** (hit target τ): $\mathrm{SNR} = 10\log_{10}(\bar{y}^2/s^2)$.

Higher SNR is always better. Bias: SNRs conflate mean and variance; variance reduction can increase SNR even when mean drifts.

Assumptions

Outcomes are non-negative (for smaller/larger-is-better); target is appropriate; replicates exist to estimate variance.

R Implementation

```r
set.seed(2026)

# Smaller-is-better: defects per batch
defects <- matrix(rpois(20 * 3, lambda = 4), 20, 3)
snr_sb <- -10 * log10(rowMeans(defects^2))

# Larger-is-better: process yield
yield <- matrix(rnorm(20 * 3, 80, 5), 20, 3)
snr_lb <- -10 * log10(rowMeans(1 / yield^2))

# Nominal-is-best: target 50
values <- matrix(rnorm(20 * 3, 50, 3), 20, 3)
means <- rowMeans(values); sds <- apply(values, 1, sd)
snr_nom <- 10 * log10(means^2 / sds^2)

head(data.frame(snr_sb, snr_lb, snr_nom))
```

Output & Results

Per-batch SNR values under the three variants; the batch maximising each SNR has the desirable combination of mean and variance.

Interpretation

"Smaller-is-better SNR was -12.5 (batch 3) vs -14.7 (batch 10); batch 3 achieved lower mean and lower variance of defects. The larger-is-better SNR identified a different optimal batch, suggesting SNRs depend on target."

Practical Tips

- SNR combines mean and variance – convenient but conflates them. Modern practice reports mean and variance separately.
- Dual-response surface methods (Myers-Carter, Vining-Myers) extend SNR thinking by fitting separate mean and variance models.
- For nominal-is-best, ensure $\bar{y}$ hits target before interpreting SNR.
- SNR is sensitive to sample size; fewer replicates inflate or deflate the metric.
- Report SNR in dB (decibels, i.e. using $10 \log_{10}$), to follow Taguchi conventions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans